

SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Code of Conduct:

I Deepak V- 192611021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Deepak.

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Improving Real-Time Video Conferencing Performance

Analysis of the Problem:

The conferencing problem is mainly caused by UDP under poor network conditions and application-level buffering issues. Video conferencing applications usually use UDP because it provides low latency.

Key Reasons:

- UDP does not retransmit lost packets
- High packet loss affects voice and video quality.
- Excessive jitter causes irregular packet arrival.

Justification:

Real-time multimedia applications prioritize low delay over perfect reliability. TCP retransmits lost packets, increasing delay, whereas UDP delivers packets immediately with acknowledgments.

Communication Requirements:

- Low latency
- Minimal jitter
- Continuous audio and video streaming

Improvements:

- Use RTP over UDP for real-time media transmission
- Enable QoS for voice and video traffic
- Increase adaptive jitter buffer size
- Enable adaptive bitrate streaming

2) Optimizing DNS Infrastructure for Faster Web Service Access

Analysis:

The organization's high DNS lookup time (800ms) is mainly caused by an inefficient DNS infrastructure. A centralized DNS system increases query latency because requests travel longer distances.

Reasons:

- Centralized DNS server handling all requests.
- High network latency between users and DNS servers.
- Inefficient DNS caching mechanism.

Proposed Architecture:

To reduce DNS lookup time below 50ms, implement,

- ~~Centralized~~ Deploy Anycast DNS with multiple global DNS servers.
- Enable DNS pre-fetching for frequently accessed domains.
- Optimize TTL values for faster cache updates.

Cons Limitations:

- Higher deployment and maintenance costs.
- Requires synchronization among DNS servers.
- TTL configuration requires management.

3) Designing a Distributed DNS Architecture for Cloud Services.

Impact:

The cloud service provider experiences high DNS lookup time (300 ms) because all DNS requests are handled by a single primary DNS server. This centralized approach creates a bottleneck during promotional events, increasing response time and reducing application performance.

Effects:

- High DNS lookup latency
- Single point of failure
- Increased server overhead
- Slow application access for users.
- Reduced service availability during heavy traffic.

A distributed DNS architecture should be implemented. to improve performance and ensure high availability

Improvements:

- Deploy Anycast DNS across multiple locations
- Enable DNS caching on local DNS servers
- Optimize TTL values for faster updates
- Use load balancing between

4) Reducing TCP Latency in High-Frequency Stock Trading Systems:

Analysis:

The increase in order acknowledgement latency from 1 ms to 40 ms is mainly caused by TCP-related issues such as flow control, Nagle's algorithm, and SYN queue limitation

These factors increase transmission delays and reduce responsiveness.

TCP - Related Factors:

- Flow control limits data transmission during traffic
- Nagle's algorithm delays new connections
- SYN queue overflow delays transmission

Effects of Transaction Processing:

- Delayed order acknowledgements reduce trading efficiency.
- Increased latency causes slower trade confirmations
- Connection delays affect new client requests.

Improvements:

- Disable Nagle's algorithm using `TCP_NODELAY`.
- Increase SYN queue size to handle peak connections.
- Optimize TCP buffer sizes to reduce delays.
- Tune TCP retransmission timeout (RTO)